



**BHASKARACHARYA NATIONAL INSTITUTE FOR  
SPACE APPLICATIONS AND GEO-INFORMATICS**  
WEEKLY PROGRESS REPORT(22/06/2026–28/06/2026)

**WEEK 6**

| PROJECT NAME | RADIO PROPAGATION ANALYSIS AND<br>PATH PREDICTION |
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**PROJECT DESCRIPTION:**

Project focuses on simulating antenna beam propagation and predicting communication coverage.

**GROUP MEMBER:**

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**GROUPID:**

2026G265

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| 22/06/2026 | <ul style="list-style-type: none"> <li>• Studied the py3dtilers workflow and its application for Cesium 3D Tiles generation. (Garima Choudhary)</li> <li>• Installed Git, configured Python 3.12 virtual environment, and successfully installed py3dtilers after resolving compatibility issues. (Manash Prasara)</li> <li>• Verified the software setup and tested the installation environment. (Aryan kumar)</li> </ul>          |
| 23/06/2026 | <ul style="list-style-type: none"> <li>• Reviewed different py3dtilers executables and their use in GIS data conversion. (Garima Choudhary)</li> <li>• Verified py3dtilers installation, analyzed its integration with the Cesium project, and tested available executables.(Manash Prasara)</li> <li>• Assisted in validating installation outputs and organizing the project files. (Aryan kumar)</li> </ul>                       |
| 24/06/2026 | <ul style="list-style-type: none"> <li>• Assisted in testing the Cesium interface after integrating terrain and OSM Buildings. (Garima Choudhary)</li> <li>• Integrated Cesium World Terrain and OSM 3D Buildings into the existing antenna visualization application and verified the 3D environment. (Manash Prasara)</li> <li>• Performed functional testing of the updated Cesium visualization and UI. (Aryan kumar)</li> </ul> |
| 25/06/2026 | <ul style="list-style-type: none"> <li>• Explored the workflow for storing antenna information using GeoJSON. (Garima Choudhary)</li> <li>• Designed the multiple antenna placement workflow, planned GeoJSON-based antenna data storage. (Manash Prasara)</li> <li>• Assisted in testing the planned multiple antenna functionality and reviewed implementation requirements.(Aryan kumar)</li> </ul>                               |
| 26/06/2026 | Holiday(Muharram)                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 27/06/2026 | Holiday(2 <sup>nd</sup> Saturday )                                                                                                                                                                                                                                                                                                                                                                                                   |
| 28/06/2026 | Holiday(Sunday)                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                |                                                                                                                                                                                                                                                                                    |
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| Next Week Plan | <ul style="list-style-type: none"> <li>• Enhance the existing UI by adding user-friendly dropdown menus for antenna configuration and visualization options. Improve the interface layout and functionality to make the application more interactive and easier to use.</li> </ul> |
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## REFERENCE:

<https://sandcastle.cesium.com/>

<https://cesium.com/learn/cesiumjs-learn/>

<https://geojson.org/>

<https://qgis.org/en/site/>

## SCREENSHOTS:



